Homogeneous parity approximants have a strict local repair cost

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

We study a discrete cusp-kernel model of local repairs to a globally margin-separated prediction field. Repairs may depend on every active coordinate and may change their signs arbitrarily many times before returning to the background signs. We prove that a homogeneous polynomial of sufficiently small degree, with parity error at most 0.34, has mean repair cost strictly above the constant-background benchmark. Amplitudes need not be bounded. The proof uses an amplitude transport and an inverse-amplitude normalization that remains valid for arbitrarily small transport rows. A more general theorem identifies the precise one-sided adjacency deficit that controls the cost. A weighted extension also covers certain affine homogeneous heads after the exact odd augmentation of the neural kernel. These are results about a fixed-background local model; they do not establish the desired sample lower bound for general adaptive neural-prior posteriors.

1 Model and result

Let $\Omega_k = \{-1,1\}^k$ have uniform probability measure, let $f(x) = \prod_{i=1}^k x_i$, and write d_H for Hamming distance. Fix $0 < \beta \le 1/2$. The local index set is $\mathcal{X} = \Omega_k \times \mathbb{Z}_{\ge 0}$. Its second coordinate is a discrete nuisance radius. For finitely supported real coefficients c_z satisfying $\sum_z c_z = 0$, put

$$V_\beta(c) = -\sum_{(x,h),(y,j)} c_{x,h} c_{y,j} \big(d_H(x,y) + h + j\big)^\beta. \tag{1}$$

This quadratic form is nonnegative. Indeed, the identity

$$s^\beta = \frac{\beta}{\Gamma(1-\beta)} \int_0^\infty (1 - e^{-st}) t^{-1-\beta}\, dt$$

reduces the assertion to positivity of the kernels $e^{-td_H(x,y)} e^{-th} e^{-tj}$. The first factor is a product of positive semidefinite two-point kernels.

For a real correction b on $\mathcal{X}$, define its squared energy by

$$\mathcal{E}_\beta(b) = \sup_{\substack{c \text{ finitely supported,} \\ \sum c = 0 \\ V_\beta(c) > 0}} \frac{\left(\sum_z c_z b(z)\right)^2}{V_\beta(c)}. \tag{2}$$

If a zero-variance contrast has nonzero value, the energy is defined as $+\infty$. Constants have zero energy. This is the Hilbert seminorm associated with the conditionally positive kernel in (1); in

*Authors are listed in alphabetical order.

particular, every contrast obeys

$$\left|\sum_z c_z b(z)\right|^2 \le \mathcal{E}_\beta(b) V_\beta(c). \tag{3}$$

Fix a background $P : \Omega_k \to \mathbb{R}$ with $|P| \ge 1$. Set $g = \operatorname{sgn} P$, $B = |P|$, $E = \{g \ne f\}$, $C = E^c$, and $R = \Pr(E)$. For each $u \in E$, an admissible repair is a correction b_u for which

1. $Z_u(x,h) = P(x) + b_u(x,h)$ satisfies $|Z_u(x,h)| \ge 1$ everywhere;
2. $\operatorname{sgn} Z_u(u,0) = f(u)$;
3. for every x, $\operatorname{sgn} Z_u(x,h) = g(x)$ for all sufficiently large h.

No condition is imposed on intermediate signs. Write $e_u = \mathcal{E}_\beta(b_u)$ and

$$v_0 = 2 - 2^\beta, \qquad C_0 = \frac{4}{v_0}, \qquad T = \frac{11}{3}, \qquad V = 2^\beta(2 - 2^\beta) + \frac{2^{2\beta} - 2^\beta}{24}.$$

All averages below use the original uniform measure, so $\mathbb{E}[\mathbf{1}_E e_u]$ includes the error probability.

For a correct vertex v, define

$$H_v = \frac{1}{k} \sum_{u \sim v,\ u \in E} B_u, \qquad \delta_v = \left(1 - \frac{H_v}{B_v}\right)_+, \qquad D_C = \mathbb{E}[\mathbf{1}_C \delta_v]. \tag{4}$$

Define also the error-to-correct edge mass and amplitude-ratio sum

$$F = \mathbb{E}\left[\mathbf{1}_E \frac{\#\{v \sim u : v \in C\}}{k}\right], \qquad M = \mathbb{E}[\mathbf{1}_C H_v / B_v].$$

Theorem 1 (Signed surplus and one-sided deficit)**.** *For $k \ge 2400$, every background and every family of admissible repairs as above satisfy*

$$\mathbb{E}[\mathbf{1}_E e_u] \ge 2T(2R + M - F) - T^2 V R \ge 2T(1 - D_C) - T^2 V R. \tag{5}$$

There is no polynomial-degree or amplitude hypothesis in this statement.

Corollary 2 (Homogeneous low degree)**.** *Suppose P is supported on a single Walsh level r, with $r/k \le 1/5$ and $|P| \ge 1$. If $k \ge 2400$ and $R \le 17/50$, then*

$$\mathbb{E}[\mathbf{1}_E e_u] \ge \frac{C_0}{2} + \frac{2}{5}. \tag{6}$$

The constants hold uniformly for $0 < \beta \le 1/2$.

The benchmark $C_0/2$ is exact for a constant background. To see this, take $P = -1$, and at each wrong center let $s = d_H(u,x) + h$ and

$$r_1(s) = (s+1)^\beta - s^\beta, \qquad b_u(x,h) = 2 - \frac{2}{v_0} + \frac{2}{v_0} r_1(s).$$

At the center $b_u = 2$, while for integer $s \ge 1$ monotonicity of r_1 gives $b_u \le 0$. Thus all required margins and eventual signs hold. The function r_1 is the representer of the contrast between $(u,0)$ and $(u,1)$ and has energy v_0. Hence this repair has energy C_0. The constant background errs on half the cube.

2 A dual inequality allowing all sign changes

We first establish the geometric estimate used in the transport proof.

Lemma 3 (Bounded neighbor dual)**.** *Fix a wrong center u. Let $w_v \ge 0$ be supported on correct neighbors of u, with $a = \sum_v w_v \le 1$ and $s = \sum_v w_v^2 \le 1/12$. Put*

$$L_w = \sum_v w_v(2 + B_u - B_v), \qquad F_w = 2^\beta(2 - 2^\beta)a^2 + (2^{2\beta} - 2^\beta)s.$$

Every admissible repair satisfies $e_u \ge C_0$ and

$$e_u \ge 2tL_w - t^2F_w \qquad (0 \le t \le 11/3). \tag{7}$$

Proof. At a fixed active point, the contrast between radii h and $h+1$ has variance

$$v_h = 2(2h+1)^\beta - (2h)^\beta - (2h+2)^\beta = \beta(1-\beta)\int_0^1\int_0^1 (2h+s+t)^{\beta-2}\,ds\,dt.$$

This decreases with h. An adjacent-radius sign change has magnitude at least 2, so it costs at least $4/v_h$. The corrected center must eventually return to its background sign, proving $e_u \ge 4/v_0 = C_0$. For $h \ge 1$,

$$v_h \le \tfrac{1}{4}(2h)^{-3/2}, \qquad v_h \le v_1 \le 2\sqrt{3} - \sqrt{2} - 2 < \tfrac{1}{20}. \tag{8}$$

The second bound uses that v_1 increases with β on $(0, 1/2]$: in its integral representation both $\beta(1-\beta)$ and $(2+s+t)^{\beta-2}$ increase with β.

Correct neighbors of a wrong parity center have the same background sign as the center. If their signs have returned to that background sign by radius H, the contrast with coefficient a at $(u, 0)$ and coefficients $-w_v$ at (v, H) has value of magnitude at least $(L_w)_+$. Direct substitution in (1) gives its variance

$$F_H = v_0a^2(H+1)^\beta + s\big((2H+2)^\beta - (2H)^\beta\big). \tag{9}$$

Thus $(L_w)_+^2 \le e_uF_H$.

If $e_u < 80$, (8) excludes every sign change at radius $h \ge 1$. Eventual anchoring therefore fixes all signs already at $H = 1$. Since $F_1 = F_w$, completing the square proves (7), in this case for every $t \ge 0$.

If $e_u \ge 80$, choose

$$H = \left\lfloor \tfrac{1}{2}(e_u/16)^{2/3} \right\rfloor + 1.$$

The first bound in (8) excludes every sign change at or beyond H. Moreover, $v_0 2^\beta \le 1$, $\beta \le 1/2$, and $s \le 1/12$ give

$$F_H \le \sqrt{(H+1)/2} + \frac{1}{12} \le \sqrt{1 + \tfrac{1}{4}(e_u/16)^{2/3}} + \frac{1}{12} \le \frac{e_u}{4(11/3)^2}.$$

For the last inequality, the ratio of the left side to e_u decreases with e_u. At $e_u = 80$, its numerator is less than $17/12$, whereas $80/[4(11/3)^2] = 180/121 > 17/12$. Consequently $(L_w)_+ \le e_u/[2(11/3)]$. This implies $e_u \ge 2tL_w \ge 2tL_w - t^2F_w$ for the asserted range of t. □

3 Amplitude transport with no discarded rows

Proof of Theorem 1. On an error-to-correct neighbor edge define

$$\mu_{uv} = \frac{B_u}{k}. \tag{10}$$

This raw transport gives

$$\sum_v \mu_{uv} \le B_u, \qquad \sum_u \mu_{uv} = H_v, \qquad 0 \le \mu_{uv} \le B_u/k.$$

Set

$$\theta_u = \frac{\sum_v \mu_{uv}}{B_u} \le 1, \quad w^0_{uv} = \frac{\mu_{uv}}{B_u B_v}, \quad a_u = \sum_v w^0_{uv}, \quad J_u = B_u a_u - \theta_u.$$

Here θ_u is exactly the fraction of correct neighbors of u. Since $B_v \ge 1$, we have $a_u \le 1$ and $w^0_{uv} \le 1/k$. Zero rows are allowed and have $a_u = \theta_u = J_u = 0$.

Let $\eta = 1/100$, $s_u = \max(a_u, \eta)$, and apply Lemma 3 to $w_{uv} = w^0_{uv}/s_u$. Its row sum is at most 1, and

$$\sum_v w^2_{uv} \le \frac{a_u}{k s_u^2} \le \frac{1}{k\eta} \le \frac{1}{24}.$$

Its linear contrast is $(J_u + 2a_u)/s_u$ and its variance is at most V. Taking $t = T$ yields

$$J_u \le s_u A_u - 2a_u, \qquad A_u = \frac{e_u + T^2 V}{2T}.$$

The baseline bound $e_u \ge C_0$, together with $C_0 \ge 4$ and $V \ge 2\sqrt{2} - 2$, implies

$$A_u \ge \frac{4 + (121/9)(2\sqrt{2} - 2)}{22/3} > \frac{200}{99}.$$

In particular $A_u - 2 \ge 0$ and $\eta A_u \le A_u - 2$. If $a_u \ge \eta$, then $J_u \le a_u(A_u - 2) \le A_u - 2$; if $a_u < \eta$, then $J_u \le \eta A_u - 2a_u \le A_u - 2$. Thus every row, including a zero or arbitrarily small row, obeys

$$J_u \le \frac{e_u + T^2 V}{2T} - 2. \tag{11}$$

The inverse-amplitude weights now turn the transport into a count identity:

$$\mathbb{E}[\mathbf{1}_E B_u a_u] = M, \qquad \mathbb{E}[\mathbf{1}_E \theta_u] = F.$$

Thus $\mathbb{E}[\mathbf{1}_E J_u] = M - F$. Averaging (11) proves the first bound in (5). The second follows from $M \ge 1 - R - D_C$ and $F \le R$. In particular, surplus incoming amplitude at correct vertices has not been discarded. □

4 Homogeneity and the explicit strict gap

Let A be normalized cube adjacency. A homogeneous Walsh polynomial of degree r satisfies $AP = (1 - 2r/k)P$. Multiplication by full parity reverses every edge, so $q = fP$ satisfies $Aq = -(1 - 2r/k)q$. At a correct vertex v, therefore,

$$H_v = (1 - 2r/k)B_v + \frac{1}{k} \sum_{w \sim v,\ w \in C} B_w \ge (1 - 2r/k)B_v.$$

This already gives $D_C \le (2r/k)(1-R)$, but retaining the correct–correct edges gives a substantially stronger bound. Pairing the two orientations of every such edge and using $a/b + b/a \ge 2$ yields

$$M = (1-2r/k)(1-R) + \mathbb{E}\left[\mathbf{1}_C \frac{A_C B}{B}\right] \ge (2-2r/k)(1-R) - F.$$

Since $F \le R$, Theorem 1 gives

$$\mathbb{E}[\mathbf{1}_E e_u] \ge 4T(1-r/k)(1-R) - T^2 V R. \tag{12}$$

For completeness, the numerical gap in Corollary 2 can be checked algebraically. The right side of (12) decreases with both r/k and R. At $r/k = 1/5$ and $R = 17/50$, its first term is $968/125$, exceeding $2T = 22/3$ by $154/375$. Put $v = 2 - 2^\beta \in [2-\sqrt{2}, 1)$. Multiplying the excess of $2T - T^2 V(17/50)$ above $C_0/2$ by v gives

$$F_0(v) = \frac{47311}{10800} v^3 - \frac{2057}{240} v^2 + \frac{37543}{5400} v - 2.$$

The derivative has positive leading coefficient and discriminant $-696300671/9720000 < 0$, so F_0 is increasing. In particular

$$F_0(v) \ge F_0(2-\sqrt{2}) = \frac{86567}{1800} - \frac{18359\sqrt{2}}{540} > \frac{121}{10000}.$$

Because $v \le 1$, the unscaled gap is at least this large. Therefore the right side of (12) exceeds $C_0/2$ by more than $121/10000 + 154/375 > 2/5$, proving the corollary.

For a mixed-degree head, the same calculation identifies a sufficient relative generator condition. With $\Delta = \sum_i (I - \mathrm{flip}_i)/2$, at correct vertices

$$H_v = B_v - \frac{2}{k}\operatorname{sgn}(P_v)\Delta P(v) + \frac{1}{k}\sum_{w\sim v,\ w\in C} B_w.$$

Consequently

$$D_C \le \mathbb{E}\left[\mathbf{1}_C \min\left\{1, \left(\frac{2\Delta P}{kP}\right)_+\right\}\right].$$

An ordinary norm bound on ΔP does not automatically control this relative quantity. Extending the theorem to arbitrary small-degree heads, and transferring an appropriate extension to the actual adaptive posterior, remain separate open steps.

5 Weighted homogeneity and two-level extensions

The argument uses a reversible nearest-neighbor walk, rather than any special property of equal coordinate weights. Let

$$A_w = \sum_{i=1}^n w_i \,\mathrm{flip}_i, \qquad w_i > 0, \qquad \sum_i w_i = 1.$$

Theorem 4. *Suppose a margin-one background H on Ω_n satisfies $A_w H = (1-2a)H$, where $a \le 1/5$ and $\max_i w_i \le 1/2400$. If its full-parity error is at most $17/50$, then every admissible repair family in the model of Section 1 satisfies*

$$\mathbb{E}[\mathbf{1}_E e_u] \ge C_0/2 + 2/5.$$

Proof. Use the raw transport $\mu_{uv} = w_i B_u$ on an error–correct edge in coordinate i. Its inverse-amplitude weight is w_i/B_v. After the same row normalization $s_u = \max(a_u, 1/100)$ used above,

$$\sum_v w_{uv}^2 \le 100 \max_i w_i \le 1/24.$$

Thus the pointwise bound (11), including its treatment of zero rows, holds unchanged. Write $F = \mathbb{E}[\mathbf{1}_E A_w \mathbf{1}_C] \le R$ and $M = \mathbb{E}[\mathbf{1}_C A_{w,E} B/B]$. Full parity conjugates A_w to $-A_w$, so on correct vertices

$$A_{w,E} B = (1-2a)B + A_{w,C} B.$$

Reversibility and $x/y + y/x \ge 2$ give $\mathbb{E}[\mathbf{1}_C A_{w,C} B/B] \ge 1 - R - F$. Consequently

$$2R + M - F \ge 2(1-a)(1-R) + 2(R-F) \ge 2(1-a)(1-R).$$

Averaging the pointwise bound therefore gives $\mathbb{E}[\mathbf{1}_E e_u] \ge 4T(1-a)(1-R) - T^2 V R$. The numerical estimate in the preceding section applies verbatim. □

There is a useful affine example, provided the signed antipodal sheet of the kernel is retained. Let k be even, $r \ge 1$ odd, and

$$P(x) = c + Q_r(x), \qquad H(x,b) = cb + Q_r(x), \qquad F(x,b) = f(x)b,$$

where Q_r is homogeneous of degree r. Then $H(x,1) = P(x)$ and $H(x,-1) = -P(-x)$. Since k is even, the two slices have exactly the same parity risk. Give each x-coordinate weight $1/(k+r)$ and the bias coordinate weight $r/(k+r)$. Every monomial of H has weighted degree $a = r/(k+r)$.

Corollary 5. *Suppose k is even, r is odd, $r/k \le 1/2399$, and $P = c + Q_r$ has margin one and parity error at most $17/50$. In the odd augmented cusp-cell model, its mean wrong-center repair energy is at least $C_0/2 + 2/5$.*

Proof. The lift H has the same margin and risk, and $a = \max_i w_i \le 1/2400$. Apply Theorem 4. □

The same construction also includes two nonconstant homogeneous levels.

Corollary 6. *Let k be even, let $r < s$ with r even and s odd, and suppose $P = P_r + P_s$ has margin one and parity error at most $17/50$. If*

$$\frac{s}{k+s-r} \le \frac{1}{5}, \qquad \frac{s-r}{k+s-r} \le \frac{1}{2400},$$

then the conclusion of Corollary 5 holds for P. In particular, adjacent levels $r, r+1$ are covered whenever $k+1 \ge 2400$ and $(r+1)/(k+1) \le 1/5$.

Proof. The odd extension is $H(x,b) = bP_r(x) + P_s(x)$ and again preserves risk because k is even. Assign the ordinary coordinates weight $1/(k+s-r)$ and the bias coordinate weight $(s-r)/(k+s-r)$. Both homogeneous components then have weighted degree $a = s/(k+s-r)$. The stated assumptions are exactly those of Theorem 4. □

Here the geometry in the corollary has a precise kernel meaning. For an odd positive definite profile F_L, consider

$$K_d(y,z) = F_L\left(\frac{1 + y \cdot z}{d+1}\right).$$

It is the $b = 1$ slice of the augmented kernel $\widetilde{K}(v, w) = F_L(v \cdot w/(d+1))$ on Ω_{d+1}. In its canonical feature space, oddness gives $\Phi(-v) = -\Phi(v)$. The slice contains one representative from every antipodal pair, so restriction and the odd extension

$$\widetilde{z}(x, b) = b\, z(bx)$$

are exactly isometric. Global margin constraints are equivalent on the two spaces. For either background above, its extension is precisely the displayed H and remains independent of all nuisance coordinates.

A bias flip is therefore a genuine distance-one edge in the augmented kernel. In original coordinates it is an antipodal input together with a sign reversal; the odd kernel's endpoint cusp is the same local cusp after this reversal. The corollary applies when the cusp-cell reduction is performed on this augmented space, with the same radial anchoring assumption as in Section 1. It does not identify an ordinary antipodal pair with adjacent points in a one-sheet Hamming model. Nor does it assert that the adaptive posterior head belongs to either two-level class.